

The Effect of Unemployment on Food Insecurity.

Abstract:

In this research paper, we tackle the problem of food insecurity and how it impacts individuals' lives. The goal is to find a key factor that increases levels of food insecurity by population to focus on combating that issue and its effects. Previous literature research has concluded that unemployment could have a leading impact on food insecurity because not having a stable source of income lowers your range of nutrition choices (M.D. Hurd & S. Rohweder, 2016). All data came from official or reputable surveys of state-level populations and was carefully handpicked to make sure no variables were inaccurate or misrepresented. The empirical analysis shows that our main independent variable, unemployment, was not significant in food insecurity. The p-value test was 0.758 for the unemployment coefficient, which is greater than our target of $P < 0.05$. That being said, this experiment instead found other factors that directly influence food insecurity. The empirical modeling found that food stamps per household per 1,000 persons have a direct relationship with decreasing food insecurity by approximately 0.97 per 1,000 population ($P < 0.05$). Furthermore, our results show a strong association between poverty rates by ethnicity and increasing food insecurity by 1.33 per 1,000 population ($P < 0.05$). Although our main experiment did not result as expected, different unexpected results underscore the interplay between socioeconomic factors and food insecurity. With this newfound outcome, different policies can be put in place to assist the impoverished, which would ultimately lead to a decrease in food insecurity. Limitations of the regression model should also be accounted for.

Introduction:

Every living organism needs to consume food to be nourished to survive. Humans, animals, plants, and even bacteria all require food as their source of energy. Every lifeform on Earth shares the same necessity of satisfying its hunger. What can vary is the type of food or quantity of food that is consumed. As of 2022, 17 million U.S. households reported that they faced some level of food insecurity (US Department of Agriculture, Key statistics, n.d). The US Department of Agriculture defines food insecurity as “a household-level economic and social condition of limited or uncertain access to adequate food” (US Department of Agriculture, Definition, n.d). Individuals are unsure if they need to reduce their quantity intake or where their next meal is coming from due to limited resources and accessibility to food. Even in America, the citizens can not escape food insecurity. *Feeding America* reports that there are around 33 million people who are food insecure in the US as of 2021 (Feeding America, 2021). Food insecurity is not a trivial issue, and understanding its participants can help prevent it. To mitigate these concerns, social economists study these matters to investigate the origin of food insecurity and what type of people go through it. This paper will determine how unemployment affects food insecurity at the state level in America.

To understand food insecurity, one must take a closer look at what food insecurity means. There are two types of food insecurity: low food security and very low food security. Low food security is defined as the reduction in the quality or variety of a diet (US Department of Agriculture, Definition, n.d). Very low food security is the disruption of eating patterns because of a reduction in food consumption(US Department of Agriculture, Definition, n.d). Food insecurity and hunger are not the same terms. Being hungry is the physiological signal that the body needs to be fueled, and can be a result of food insecurity. On the contrary, identifying as having high food security means that there was/is no problem accessing food.

To many people, their next meal is not a worry of where it is coming from but rather what type of food they will be eating. Many Americans have access to a wide variety of cuisines at their fingertips. Those who are not as fortunate often struggle to even find a meal or must skip their next meal. This limited access to food can lead to a wide variety of negative health effects. Without proper nutrition, the overall quality of life can decrease. Improper diets can lead to heart disease, diabetes, and many other chronic illnesses (National Institutes of Health, n.d).

So those who are food insecure are more susceptible to these illnesses. Food insecurity adds to a health inequality that can hinder the development of future generations. It is important to research this national health crisis as it holds weight in the health and social impacts on those who go through it. With a proper understanding of food inequality, different measures can be put into place to prevent or assist those undergoing it.

One major understanding of where food insecurity originates from is income. If there is money, then there is food to buy. A 2012 paper states that individuals are more likely to be food insecure in low-income households

than in higher-income households. (Gundersen, C & Oliverira, V. 2001) There are multiple scenarios where a household could be in a low-income situation, as the economic market directly correlates with the prosperity of its participants. There is also a wide range of factors that can influence a household or individual to be classified as low-income. Education level, job security, and harsh market conditions all contribute to affecting low-income participants. But one key factor remains: living in a lower-income household means that the source of income is not as high as others. This brings in the subject of unemployment as to why one would be classified as low-income.

Once an individual loses their job, financial securities begin to crumble. The labor market is constantly changing; with new technological advancements, more and more people are prone to becoming unemployed (BLS, 2017). There are many other factors influencing unemployment, but when these changes happen, individuals find themselves struggling to provide a livable wage. Household income is hit, and it becomes difficult to afford necessities like food. Oftentimes, there are lifestyle adjustments to fit these different situations. Increasing food insecurity could be an option considered during downsizing, or individuals could find other means to combat their unemployment status.

This paper will study the effects of unemployment on food insecurity. This answer will help us understand if monetary needs have an impact on food insecurity.

Literature Review:

In recent times, there has been a multitude of studies about food insecurity, the effects of it, and who are the main participants of it. However, understanding the direct relationship between food insecurity and unemployment is essential for informing conglomerates that are willing to find solutions to fix this socioeconomic behavior. This paper will use previous research that examines what directly affects food insecurities to determine what is relevant.

A study from *Public Health Nutrition* shows that low-income households are claimed to be associated with increased food insecurity. It was reported that in 2014, people with income levels below the federal poverty level were 100% likely to be food insecure (Furness, Simon, Wold, & Asarian-Anderson, 2004).

However, *Mark Nord and C. Philip Brent* reported that 20% of households in America who reported food insecurity were of high or mid-range income levels (Nord, M & Brent, CP. 2002). It was reported that many of these instances were due to factors of uneven income or changes in residents in the household (Nord, M & Brent, C.P. 2002). There are different situations where upper-income households have the opportunity to become food insecure. Low-income households are not the only target of food insecurity.

A similar study goes into the food insecurity of rural communities. The *PMC* found that there is a disproportionate level of food insecurity in rural areas. Many of these communities have to find coping mechanisms to combat their situation (Seligman, H K, Berkowitz, S A, & Kim, J J. 2021). Depending on where they are living, it

was said that food source and resource management were small, orderly skills that were picked up to live under food insecurity. Many times, budget is a big factor when considering different food products because food is a purchase made daily. These social insights of those undergoing food insecurity struggles add to the knowledge and impact of how food security can affect an individual.

One study looks into consumption smoothing during times of unemployment, specifically for households. This study by *M.D. Hurd and S. Rohweder* looked into the effects of the 2007 economic crisis on unemployment and how households were combating consumption issues after the recession ended (M.D. Hurd & S. Rohweder. 2016). The study compared monthly household spending before unemployment and spending up to 40 months after being unemployed (M.D. Hurd & S. Rohweder. 2016). The result found that just two months after being unemployed, the average household decreased its spending by 83 percent (M.D. Hurd & S. Rohweder, 2016). That trend continued, and after about a year, spending reached a 70 percent reduction compared to pre-unemployment (M.D. Hurd & S. Rohweder, 2016). This shows that households are choosing to spend less on daily expenditure. Food consumption is a priority on that list due to its necessity, so it is not inappropriate to assume that low food insecurity could affect some of these households.

Overall, previous research has shown that income will be the deciding factor if someone identifies as food insecure or not. This leads to distinguishing the factors that employment has on food insecurity.

Empirical Model:

A linear regression model is used to estimate the effects of unemployment on food insecurity. This paper will use the following model for all calculations:

$$FdInsec = \alpha + \beta_1 Unemploy + \beta_2 FdStamPerHH + \beta_3 FdStampHHRate + \beta_4 HHIncoe + \beta_5 CostOfLiving + \beta_6 Povratebyeth + \beta_7 Region + \beta_8 Ethnicity + u$$

This model will estimate exactly how much each variable affects food insecurity. For example, the food stamps per household coefficient can indicate whether or not there would be an increase in the population that identifies as food insecure per 1,000 people. The unemployment coefficient will estimate whether the unemployment population in each state will increase or decrease food insecurity from a one-point increase in the state unemployment population. This is how the study will determine if unemployment alters the population rate of food insecurity.

Data:

The population data of people who identify as food insecure comes from *Feeding America* (Feeding America, 2022). Feeding America is a nonprofit organization whose goal is to reduce American hunger nationwide. The organization finds different ways to help provide food for those in need. Their method to collect 2021 data was based on the U.S. Department of Agriculture (USDA) food insecurity measure (Feeding America, 2022). *Feeding America* required a personalized request to obtain numerical data, but this map of data should be relevant to the data

being used in this study: <https://map.feedingamerica.org/>. Since there is no specific demographic, such as age, race, etc., to identify as food insecure, the entire reported population was taken into account. This way, all food-insecure people are accounted for properly.

The unemployment variable was from the *U.S. Bureau of Labor Statistics*. Their method of collecting 2021 state-level unemployment in America was through the Local Area Unemployment Statistics program. (Bureau of Labor Statistics. 2021) The *U.S. Census Bureau* conducts this survey for the *Bureau of Labor Statistics* (Bureau of Labor Statistics, n.d.). This data is available at: bls.gov. To be classified as unemployed, an individual has to be not working at all through the week, and the individual has to be actively searching for employment. This individual is also ready and able to be accepted back into the workforce. The 2021 Unemployment data was reported in population per 1,000 persons at the state level.

Other control data included are Food Stamps Per Household (HH), Food Stamps Household Rate, Household Income, and Cost of Living. The *World Population Review* provided food stamps per household and the Food stamp household rate. Both coefficients were measured annually at the state level in 2024. Food stamps per household use the unit of population per 1,000 persons, and the Food stamp household rate is a percentage rate. As an addition to the unemployment variable, these units were added as controls due to the condition that is required for a person to have access to food stamps. The Supplemental Nutrition Assistance Program (SNAP) has a list of different requirements to be eligible for SNAP food stamps (Texas Health and Human Services Commission, nd). Some of these requirements fall under having a limit to monthly income, the number of members in your family, and citizenship status (Texas Health and Human Services Commission, nd). Because this type of eligibility is required by one of the large food stamp distributors, it is important to hold these factors constant against unemployment. This applies the same to household income and cost of living. Household income and cost of living are control values implemented to reduce the financial demographic's influence, thereby isolating the effects solely attributed to unemployment. Household income is standardized as the median household income by the state in dollars in 2021. It can be found on the same page at [WorldPopulationReview/HouseholdFoodStaprate](https://WorldPopulationReview.com/HouseholdFoodStaprate).

The cost of living is based on an index (Worldwide Cost of Living, nd). This index bases the city it is measuring at 100. Any statistic measuring above or below the base value represents that a city is more expensive or less expensive to live in. For example, the cost of living index measures Alabama as 88.8, meaning that that is a relatively less expensive state to live in. The cost of living data was taken annually at the state level in 2024, which can be found at [WorldPopulationReview/CostOfLivingIndex](https://WorldPopulationReview.com/CostOfLivingIndex).

The poverty rate by ethnicity is based on the U.S. Census Bureau's poverty threshold measurement for a family. It claims that households with two adults and one child would be impoverished if they were making \$23,556 a year as of 2022 (Kaiser Family Foundation- Poverty Rate by Race/Ethnicity). The 2021 data was taken from [KFF.org](https://kff.org) and was compressed to people in poverty per 1,000 persons at the state level. This variable was added to

ensure that no omitted variable bias would occur. Having poverty as a solution to the omitted variable bias is made possible because it adds more collected data to mitigate any possible bias. This is also true with region and race. Both variables add a captured subgroup of data that affects the dependent variable of food insecurity. This way, there is more information about demographics and ethnicity to ensure the regression is as detailed as possible with no bias.

The data for the region was collected in dummy variable standards. If the state is in the west region of the US, then a 1 is used for data; if the state is not in the west region, then a 0 is filled in. The region's data comes from [Census.gov](https://www.census.gov). The south was removed from the regression due to collinearity, which could cause wrong interpretations of the regression coefficients.

The data for ethnicity was taken from [KFF.org/populationdistributionbyethnicity](https://www.kff.org/populationdistributionbyethnicity). This data is from 2021 and is reported with variables Black, Hispanic, and Asian, all taken in population per 1,000 persons (Kaiser Family Foundation- Distribution by Race/Ethnicity). This data was used to prevent any omitted variable bias.

Table 1 is below which represents the descriptive statistics for the coefficients.

Variable	Obs	Mean	Std. Dev.	Min	Max
FdInsec	51	704.0333	851.9909	42.29	4104.06
Unemploy	51	183.6458	258.863	8.033	1489.611
FdStampsPerHousehold	51	272.4001	293.7551	12.313	1183.873
FdStampHHRate	51	11.01569	2.698842	5.3	17
HHIncome	51	74846.33	12229.9	52658	97501
CostOfLiving	51	104.002	18.24097	85.3	179
Povratebyeth	51	811.4667	950.7473	64.2	4728.4
West	51	0.254902	0.4401426	0	1
Northeast	51	0.1764706	0.3850134	0	1
Midwest	51	0.2352941	0.4284033	0	1
BlackspPer1000	51	735.7098	893.7869	3.9	3281.1
AsianspPer1000	51	363.8941	852.5311	4.2	5811.1
HispanicsPer1000	51	1200.818	2740.876	12.6	15447.9

Table 1

Table 1 shows trends that show that food insecurity does rely on monetary factors. California has the largest population of people who identify as food insecure. Also, California not only has the largest unemployment population and largest population for food stamps per household, but is also near the top percentile for household income and cost of living, with \$91,777 and 134.5, respectively. California also has the highest Asian and Hispanic population per 1,000 persons and the highest population of impoverished people by ethnicity. This evidence leans towards the idea that food-insecure people also struggle financially with money.

Empirical Results:

To ensure the validity of our regression results, several diagnostic tests were performed. A Breusch-Pagan test revealed signs of heteroskedasticity in the residuals, indicating that the assumption of constant variance was violated. To correct for this, all regressions were re-estimated using heteroskedasticity-robust standard errors. Additionally, multicollinearity was assessed using Variance Inflation Factors (VIFs). While some variables—such as household income and food stamp rate—showed moderate collinearity, no values exceeded the commonly used threshold of 10, suggesting that multicollinearity was not a serious issue. The dramatic shift in coefficients and significance between *Tables 2* and *3* also pointed to omitted variable bias in the original model. To address this, alternative specifications were estimated by including additional controls such as poverty rate by ethnicity, region, and race. These controls significantly improved the model and changed the interpretation of key variables, reinforcing the importance of controlling for underlying demographic factors. Lastly, as a robustness check, a log-log model was estimated to test for non-linear effects, which confirmed the direction and significance of our main findings.

Table 2 below shows the results of the regression model without any attempt to fix the omitted variable bias issue. Higher rates of unemployment lead to more of the population being food insecure. When there is a one-point increase in the unemployment population, food insecurity increases by about 1.75 per 1,000 population with a p-value < 0.05. Other factors that also affect food insecurity are food stamps per household and household income. Food stamps per household also had a 1.43 per 1,000 population on food insecurity when it was increased by one point, with a p-value < 0.05.

. regress FdInsec Unemploy FdStampsPerHH FdStampHHRate HHIncome CostOfLiving

Source	SS	df	MS	Number of obs	=	51
Model	34803573.2	5	6960714.64	F(5, 45)	=	210.10
Residual	1490851.08	45	33130.0239	Prob > F	=	0.0000
				R-squared	=	0.9589
				Adj R-squared	=	0.9544
Total	36294424.3	50	725888.486	Root MSE	=	182.02

FdInsec	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
Unemploy	1.749834	.278977	6.27	0.000	1.187945	2.311722
FdStampsPerHH	1.427999	.2506361	5.70	0.000	.923192	1.932806
FdStampHHRate	-23.38481	13.35963	-1.75	0.087	-50.29249	3.522873
HHIncome	-.0081184	.0039411	-2.06	0.045	-.0160563	-.0001806
CostOfLiving	.3737076	2.396225	0.16	0.877	-4.452538	5.199953
_cons	820.0659	262.6377	3.12	0.003	291.0863	1349.045

Table 2

Household income is negatively correlated with food insecurity. When household income is increased by \$1, food insecurity decreases by 0.008 per 1,000 population with a p-value < 0.05. The food stamp household rate and cost of living did not have a low enough p-value, so those coefficients did not have a significant effect on food insecurity. So we would fail to reject the null hypothesis and would need to look into the reasons why in future research. This data is under harsh conditions of omitted variable bias, and once we add variables to combat this issue, the results change drastically.

Table 3 below shows the empirical result with added variables to reduce bias. These results are drastically different. Unemployment is no longer statistically significant enough for results to be concluded at a p-value < 0.05. What results we can conclude is that when there is a one-point increase in the food stamps per household, food insecurity decreases by about 0.97 per 1,000 population with a p-value < 0.05. We also conclude that a person is more likely to be impoverished if they are food insecure, as there is a one-point increase in the poverty rate by ethnicity per 1,000 persons, food insecurity increases by about 1.33 per 1,000 population, with a p-value < 0.05.

```
. regress FdInsec Unemploy FdStampsPerHH FdStampHHRate HHIncome CostOfLiving Povratebyeth
> per1000 West Northeast Midwest Blacksper1000 Asiansper1000 Hispanicsper1000
```

Source	SS	df	MS	Number of obs	=	51
Model	36114367.5	12	3009530.63	F(12, 38)	=	635.15
Residual	180056.745	38	4738.3354	Prob > F	=	0.0000
				R-squared	=	0.9950
				Adj R-squared	=	0.9935
Total	36294424.3	50	725888.486	Root MSE	=	68.836

FdInsec	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
Unemploy	.1108346	.3576511	0.31	0.758	-.6131923	.8348614
FdStampsPerHH	-.9702536	.2940141	-3.30	0.002	-1.565454	-.3750532
FdStampHHRate	4.02283	5.984159	0.67	0.505	-8.091467	16.13713
HHIncome	.003068	.0019023	1.61	0.115	-.000783	.006919
CostOfLiving	.0764287	1.047346	0.07	0.942	-2.043812	2.19667
Povratebyethper1000	1.326221	.1452265	9.13	0.000	1.032225	1.620216
West	15.2951	37.81476	0.40	0.688	-61.25687	91.84708
Northeast	18.81659	41.19221	0.46	0.650	-64.57268	102.2059
Midwest	15.5662	36.72851	0.42	0.674	-58.78678	89.91918
Blacksper1000	-.0491546	.0335385	-1.47	0.151	-.1170497	.0187405
Asiansper1000	-.1706232	.0492814	-3.46	0.001	-.2703883	-.0708582
Hispanicsper1000	-.0056907	.0195432	-0.29	0.772	-.0452539	.0338724
_cons	-315.8972	147.9595	-2.14	0.039	-615.4256	-16.36871

Table 3

Table 2 and *Table 3* are completely different interpretations of the data; *Table 2*'s results are reciprocal to what *Table 3* reported. *Table 3*'s data is objectively the more accurate representation of this experiment. The results

of *Table 2* and *Table 3* emphasize the importance that researchers should be extremely aware of any bias that could be possible in a regression experiment.

Diagnostic testing is important in regression analysis. This ensures validity, reliable data, and quality of statistical modeling. This model will be testing four key diagnostics tests: Heteroskedasticity, the Ramsey Test to test for misspecification of functional form in this model, Normality, and Multicollinearity.

Heteroskedasticity is where the variance of error or residual in the model is not constant across the independent variable. The spread or dispersion of the residuals is different across the range of values in the independent variables. To complete this test utilization the White Test will be utilized.

```
. estat imtest, white
```

White's test

H0: Homoskedasticity

Ha: Unrestricted heteroskedasticity

```
      chi2(50) =  51.00
Prob > chi2 = 0.4341
```

Cameron & Trivedi's decomposition of IM-test

Source	chi2	df	p
Heteroskedasticity	51.00	50	0.4341
Skewness	12.68	12	0.3924
Kurtosis	1.30	1	0.2549
Total	64.98	63	0.4075

Table 4

The White's Test, seen in *Table 4*, assesses the presence of heteroskedasticity in the model. In the test, since the p-value of 0.4341 is greater than the significance level of 0.05, we fail to reject the null hypothesis. There is not enough evidence to conclude that there is heteroskedasticity in the regression model, which suggests that the assumption of homoscedasticity, varying variance of residuals, is not violated. White's test concluded there is no evidence of heteroskedasticity in the model.

The Ramsey Regression Equation Specification Error Test (RESET) test finds potential misspecification in the regression model. This means identifying if there are any omitted variables or nonlinearities in the equation.

```
. ovtest
```

Ramsey RESET test for omitted variables

Omitted: Powers of fitted values of **FdInsec**

H0: Model has no omitted variables

F(3, 35) = 4.82

Prob > F = 0.0065

Table 5

In this case, the test in Table 5 confirms the inclusion of the fitted value in food insecurity that might have been omitted from the original regression. There was no omitted variable in the model, but since the p-value is less than the significance 0.05, we reject the null hypothesis. This implies that the model likely had omitted variables or nonlinearities, which could be fixed by adding additional variables to improve its fit.

Normality in multi-linear regression models often refers to the normality of the residuals. The residual is the difference between the dependent variable and the predicted values of the regression model. The distribution of these residuals should be normal or in a bell-curve fashion. If the residuals are not normally distributed, then this can indicate that the model is not representing the relationship in the data correctly. The model will be tested on the guidelines of the Skewness-Kurtosis All Normality Test.

```
. sktest FdInsec Unemploy FdStampsPerHH FdStampHHRate HHIncome CostOfLiving PovratebyEth West Midwest Northeast Blackspers1000 Asian  
> spers1000 Hispanicspers1000
```

Skewness and kurtosis tests for normality

Variable	Obs	Pr(skewness)	Pr(kurtosis)	Joint test	
				Adj chi2(2)	Prob>chi2
FdInsec	51	0.0000	0.0000	34.43	0.0000
Unemploy	51	0.0000	0.0000	42.46	0.0000
FdStampsPerHH	51	0.0000	0.0050	20.42	0.0000
FdStampHHRate	51	0.9287	0.6792	0.18	0.9144
HHIncome	51	0.3324	0.1658	3.02	0.2207
CostOfLiving	51	0.0000	0.0004	25.04	0.0000
PovratebyEth	51	0.0000	0.0000	32.00	0.0000
West	51	0.0017	0.1983	9.64	0.0081
Midwest	51	0.0007	0.6427	9.81	0.0074
Northeast	51	0.0000	0.1238	15.52	0.0004
Blackspers1000	51	0.0002	0.0747	13.24	0.0013
Asianspers1000	51	0.0000	0.0000	64.25	0.0000
Hispanicspers1000	51	0.0000	0.0000	50.59	0.0000

Table 6

Most of the variables have strong evidence against normality in *Table 6*. Food insecurity, unemployment, food stamps per household, cost of living, poverty rate by ethnicity, and each ethnicity all have a very low p-value, which suggests against normality. Food stamp household rate and household income have high joint test p-values, which suggests that they are closer to being considered normal compared to the other values. Most of the variables in the dataset greatly deviate from the normal distribution based on the test. This can impact the validity of statistical analyses of the model.

Multicollinearity is when two or more predictor variables can be linearly predicted from another predictor variable with a significant level of accuracy. This causes problems of unreliable coefficients, where it is difficult to determine the effect of each predictor variable. This can also lead to a wider range of confidence intervals, so the estimate loses its precision. In this model, we will use the Variance Inflation Factor Test (VIF) to test for multicollinearity.

. vif

Variable	VIF	1/VIF
PovratebyEth	201.17	0.004971
Unemploy	90.45	0.011056
FdStampsPe~H	78.71	0.012704
Hispani~1000	30.28	0.033028
Asiansp~1000	18.63	0.053687
Blacksp~1000	9.48	0.105463
HHIncome	5.71	0.175086
CostOfLiving	3.85	0.259645
West	2.92	0.342095
FdStampHHR~e	2.75	0.363324
Northeast	2.65	0.376768
Midwest	2.61	0.382775
Mean VIF	37.44	

Table 7

This regression model has a large case of multicollinearity, shown in *Table 7*. The standard variance inflation factor is worrying for any value greater than 10. We see that poverty rate by ethnicity, Unemployment, Food stamps per household, and Hispanic and Asian ethnicity all impact the model, suggesting high multicollinearity. This clutters our independent variable's effect on our dependent variable, making it hard to determine the precise effect of the model.

The diagnostic test provided insight into the strengths and weaknesses of the regression model. Heteroskedasticity was not found according to the White Test, implying that the variance of the residual remains constant across the board. RESET shows that there are potential misspecifications in the model that indicate the presence of variables or nonlinearities. Additional variables could be added to test the null hypothesis to improve the accuracy of the model. The Skewness Normality evaluation showed many deviations from the majority of the

variables. Food insecurity, unemployment, food stamps per household, cost of living, poverty rate by ethnicity, and each ethnicity were all tested against normality. Different transformations could allow the data points to be distributed in a more normal fashion. This would ensure the robustness of the model's performance. Lastly, in the Multicollinearity examination, variables: poverty rate by ethnicity, unemployment, food stamps per household, and Hispanic and Asian ethnicity all demonstrate significant issues of multicollinearity. Again, using transformation would enhance the data and result in a more accurate estimation of the impact of the variables. Some common solutions to address the issues of this regression could be: identifying omitted variables or nonlinear relationships, using transformation of the variable to fix heteroskedasticity and normality, and finding uncorrelated variables to mitigate multicollinearity. There are many flaws in the data that should be considered to address the validity and reliability of the analysis. It is important to test and interpret these flaws to have a deeper understanding of the complexities of multilinear regression models to better utilize the model in the future.

Conclusion:

The study shows that unemployment is not a key factor in determining food insecurity. This is a surprising discovery as the hypothesis, after literature research, concluded that unemployment would affect food insecurity (M.D. Hurd & S. Rohweder, 2016). The result concluded otherwise. This coincides with the study done by *Public Health Nutrition*, where the results were that low-income households under the poverty line are more likely to be food insecure (Furness, Simon, Wold, & Asarian-Anderson, 2004). Different social welfare program policies should target these individuals in trying to assist them when considering their preferred food choices. Policy changes include loosening the SANP requirements, education on proper budgeting strategies, and a focus on reducing the poverty rate. This also coincides with the experiment because, as was shown, food stamps help to decrease those who identify as food insecure. The results should be used in a manner to fix this socioeconomic issue by combating the root of the issue.

The results of this study should not be taken as perfect. Much of the data used was collected by a survey. There are many flaws to this method, such as sampling bias, questionable wording, and response fatigue. Each of these flaws has the potential to provide inaccurate data to the survey. In addition, this data only passed two out of four tests, which concluded that much of the model had data that was contaminating the accuracy of the model. Further research should go into the selection and adjustment of the variable being compared to ensure that the model displays the most accurate interpretation of the model.

This study also looked at how omitted variable bias is a key factor to take into consideration when participating in empirical modeling experiments. It could be a bit limiting fact that provides inaccurate data and a misleading representation of the research.

In addition to the challenges faced with food insecurity, it is important to recognize the multifaceted nature of the factors that influence food insecurity. There are also a multitude of factors that influence food insecurity,

which are not just limited to the ones provided. However, acknowledging these complex social issues is crucial for creating interventions and policies to address these issues.

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